Customer Shopping Behavior Analysis

3,900 transactions across 18 behavioral and demographic features.

DATA REPORT

Retail Analytics

End-to-end data pipeline from raw logs to commercial insight.



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DATA SCOPE

3,900 purchases, 18 features, one question: where does growth come from?

Core dimensions: Demographics, purchase values, and product preferences tracked across categories.

Behavioral metrics: Subscription status, discount application, and previous purchase cadence.

Data quality note: 37 missing Review Rating values identified during initial exploratory data checks.

Dataset scope · 3,900 records · 18 variables captured per transaction



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ARCHITECTURE

The pipeline: Python cleaning, SQL analysis, Power BI

01

01 · Ingestion & Imputation

Pandas load, null audit, and median rating imputation computed per category.

02

02 · Feature Engineering

Snake case standardization; engineered age_group brackets and purchase_frequency_days.

03

03 · Feature Selection

Dropped promo_code_used as redundant with discount_applied after correlation test.

04

04 · Database & BI

Validated datasets loaded to SQL warehouse, connected directly to Power BI reports.

End-to-end data pipeline · Python (Pandas) → SQL → Power BI

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BASELINE METRICS

The typical purchase: $59.76 average, 3.75 average rating

$59.76

Average transaction amount across all purchases

Spend spans from $20 to $100

3.75

Average review rating post-imputation

Out of 5.0 maximum rating

44 yrs

Average customer age across transactions

Customer age spans 18 to 70

25.4

Average previous purchases per buyer

Reflects high repeat purchase frequency

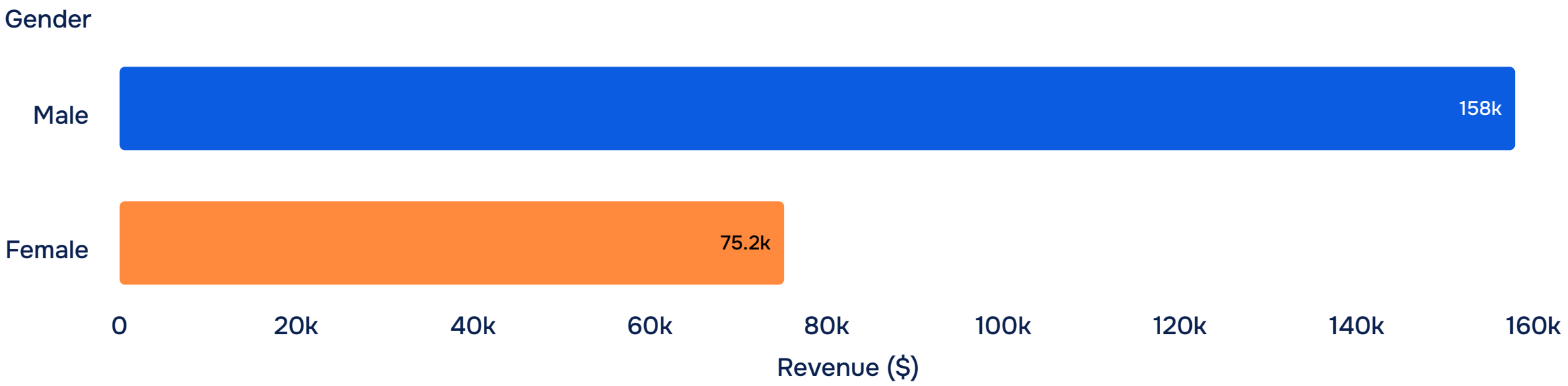
Descriptive statistics · Full customer cohort summary (N = 3,900)

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2.1×

GENDER ANALYSIS

Male customers generate about twice the revenue of female customers

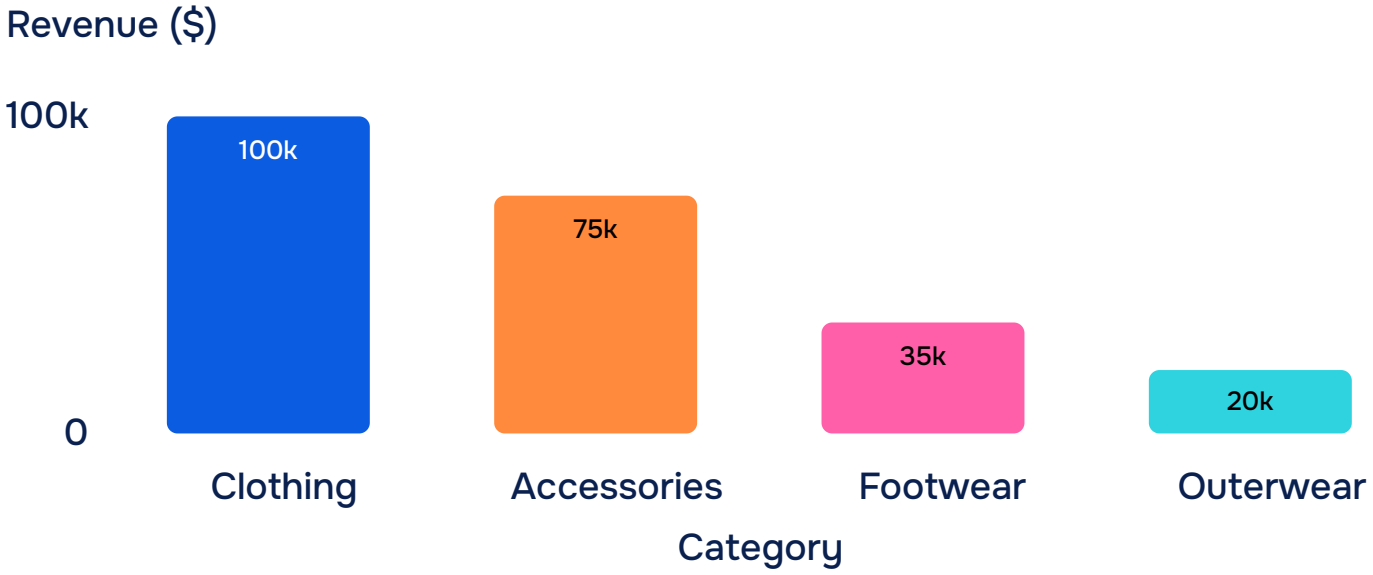


Total gross revenue split by customer gender · Male: $157,890 vs Female: $75,191

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CATEGORY BREAKDOWN

Clothing and accessories carry the revenue



Gross revenue contribution by product category

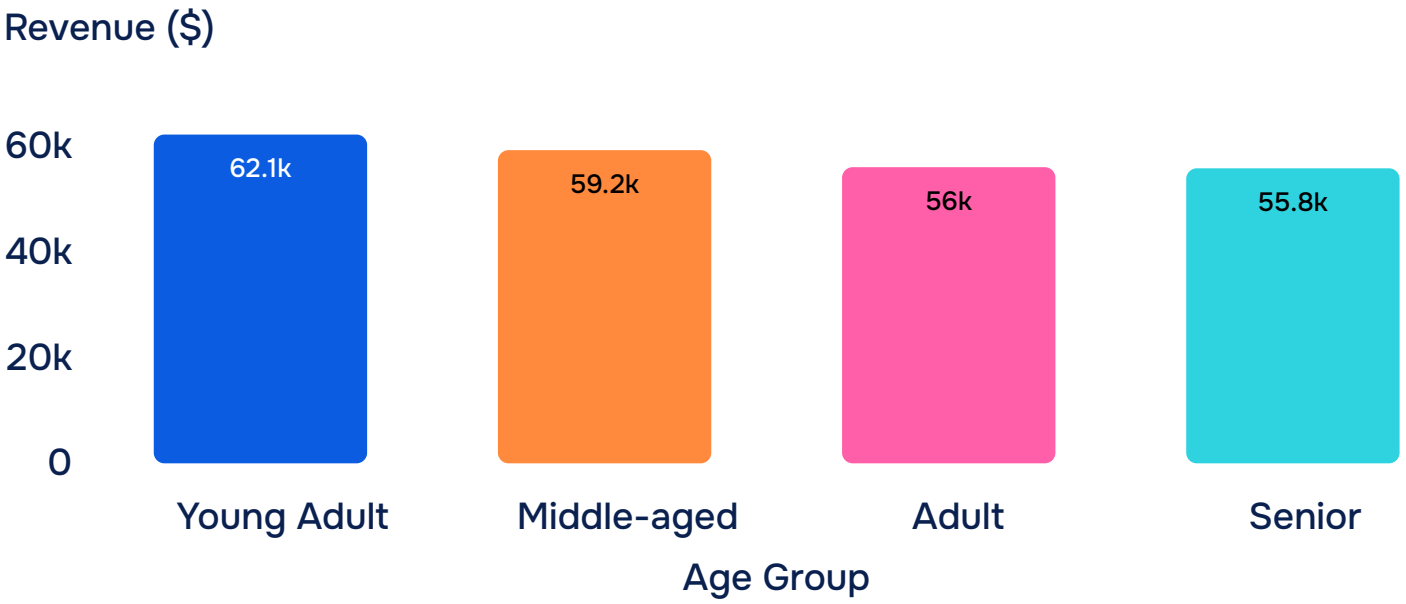
Top Volume Drivers

Top items: Jewelry and Blouse (171 orders each), Sandals, Jacket

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DEMOGRAPHIC SPREAD

Revenue is spread evenly across age groups



Revenue distribution by engineered age bracket (range 18–70)

Balanced

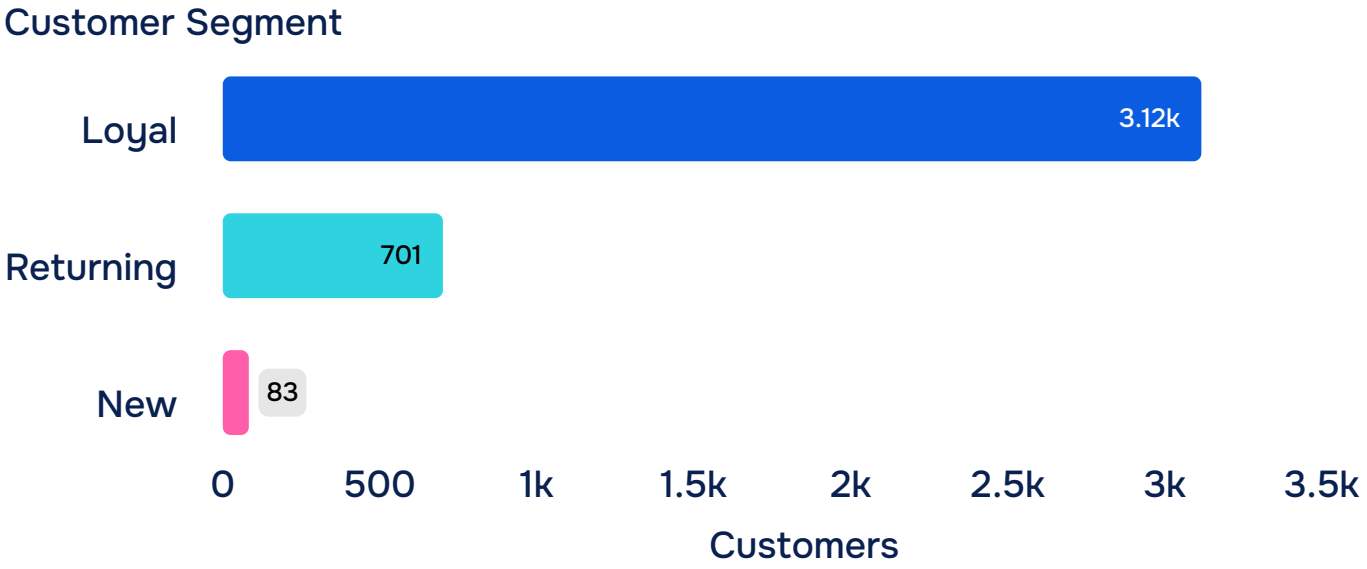
Revenue delta between Young Adult ($62,143) and Senior ($55,763) is under 11%.

No single age cohort dominates demand, indicating broad catalog appeal.

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RETENTION & COHORTS

Loyal customers make up 3,116 of 3,900



Customer volume split across maturity lifecycle segments

CHALLENGE

Thin Acquisition Pipeline

With 3,116 loyal and 701 returning shoppers, the customer base is highly mature.

Only 83 new customers were acquired, signaling that future growth requires opening top-of-funnel acquisition.

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MEMBERSHIP IMPACT

Subscribers are 27% of customers and spend no more

Subscribers (27%)

1,053 customers enrolled

$59.49 average purchase amount

$62,645 total segment revenue

958 repeat buyers (>5 purchases)

Non-Subscribers (73%)

2,847 customers unenrolled

$59.87 average purchase amount

$170,436 total segment revenue

2,518 repeat buyers (>5 purchases)

Subscription comparison: Basket size shows zero premium ($59.49 vs $59.87).

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PROMOTION SENSITIVITY

Discounts: heavy on hats, sneakers, and coats

Item	Discount rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

839

Discount users still spent above average.

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